

Introduction to Ansible



- Simple. Powerful. Agentless Automation



Provisioning



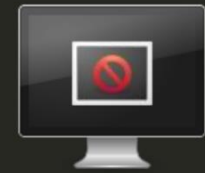
Configuration
Management



Continuous
Delivery



Application
Deployment



Security
Compliance



- Time
- Coding Skills
- Maintenance



- Simple
- Powerful
- Agentless

What Is Ansible?

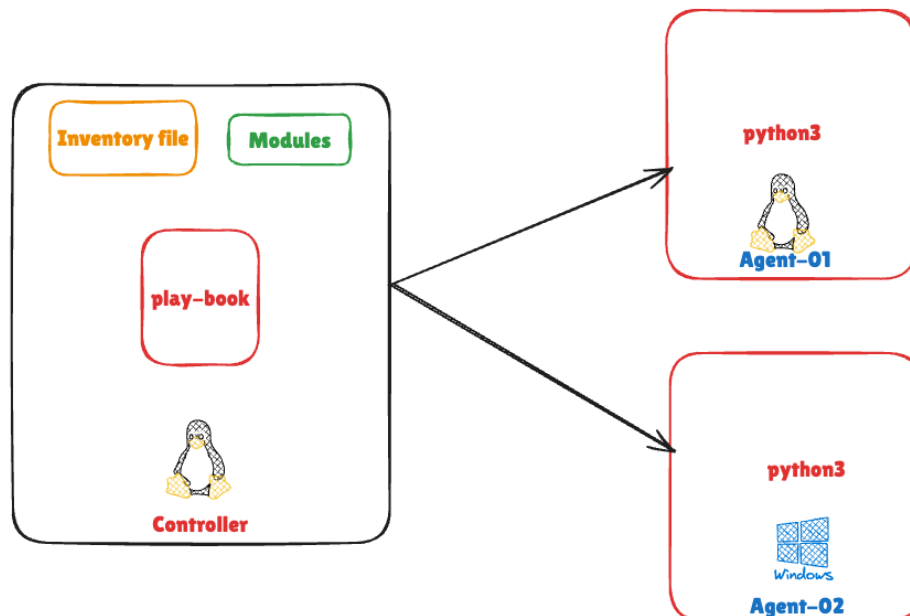
- Ansible is an open-source IT automation tool.
- Developed by Red Hat.
- Used to provision infrastructure, configure systems, deploy applications, and orchestrate workflows.

Key Features of Ansible

- Written in Python
- Uses YAML for playbooks
- Agentless – no software installed on nodes
- Uses SSH or WinRM to communicate
- Simple and human-readable syntax

Architecture

- Control Node: Runs Ansible (Linux only)
- Managed Nodes: Remote systems (Linux/Windows)
- Inventory: List of managed hosts
- Modules: Perform tasks (e.g., yum, copy, file)
- Playbooks: YAML files describing automation

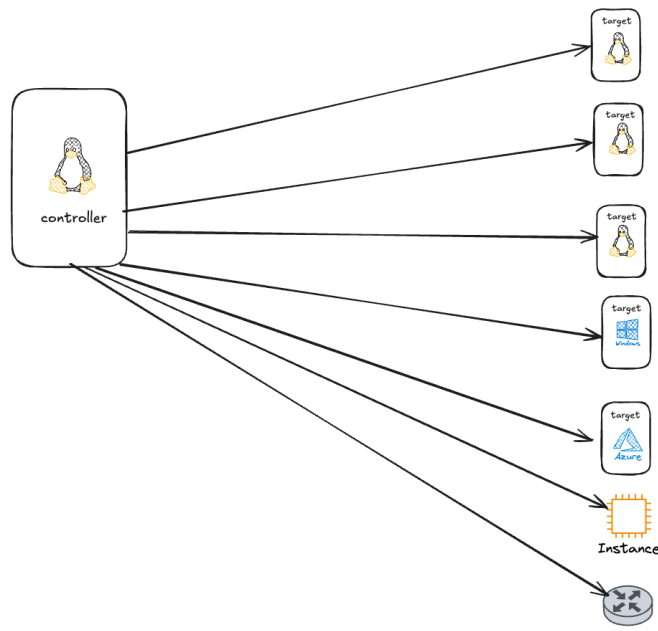


Why Use Ansible? (The Benefits)

-  Agentless – No daemons or agents on nodes
-  Simple – Easy to learn with YAML syntax
-  Powerful – Handles complex deployments and orchestrations
-  Repeatable – Ensures consistent environments
-  Idempotent – Tasks run only when needed
-  Extensible – Use custom modules, plugins, roles

Real-World Use Cases

- Configure hundreds of servers with one command
- Install apps like NGINX, Docker, Java, etc.
- Push custom configurations across environments
- Orchestrate multi-tier applications (e.g., web + DB)
- Deploy cloud infrastructure (AWS, Azure, GCP)

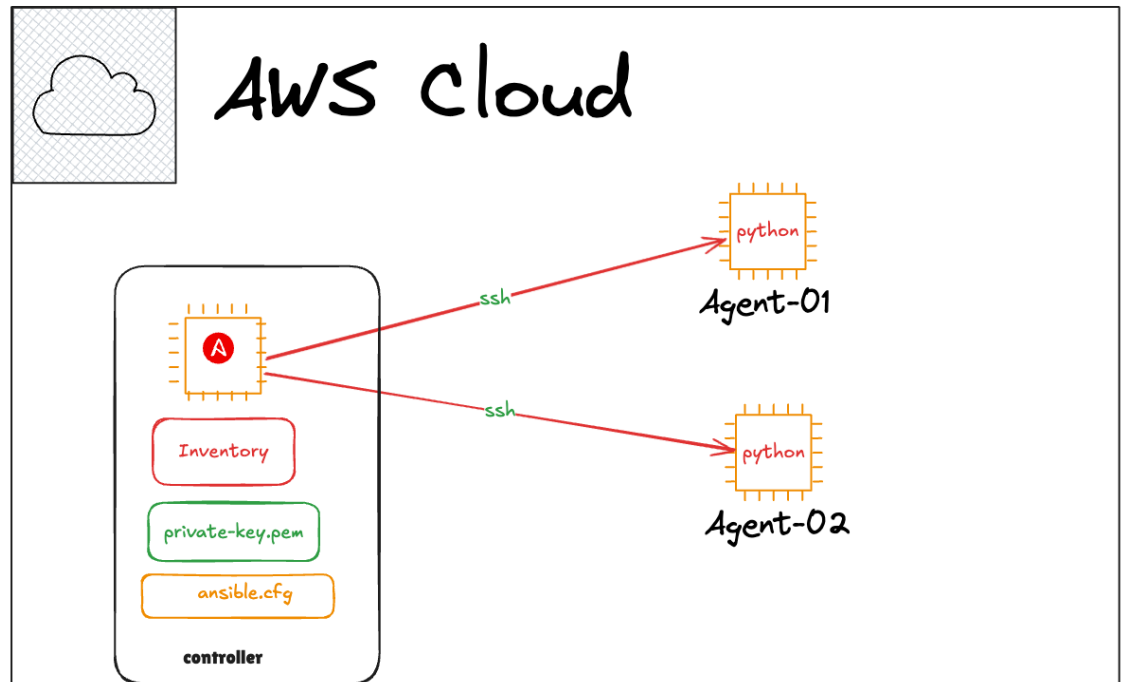


Ansible vs Traditional Scripts

- Shell Script vs Ansible:
- - Syntax: Bash vs YAML
- - Readability: Low vs High
- - Idempotent: ❌ vs ✅
- - Error Handling: Manual vs Built-in
- - Agent Required: ❌ vs ❌
- - Scalable: ❌ vs ✅

Lab Setup

LAB setup



LAB steps

- Go to the lab setps word file.